**FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4**

**PHYSICS / APPLIED PHYSICS**

**PROPERTIES OF SOLIDS**

Module IV: Dielectric and Magnetic Properties of Solids

| **Maximum Time: 2 Hours** | | **Maximum Marks: 70** |
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| | **Section A** | |
| | **All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)** | |
| 1 | Using the classical orbital picture, explain how Larmor precession produces diamagnetism. Why is the induced moment opposite to the applied field? State the approximate temperature dependence of diamagnetic susceptibility. | **3** |
| 2 | A paramagnetic material has $10^{28}$ atoms/m$^3$. Its susceptibility at $350\ \mathrm{K}$ is $2.8 \times 10^{-4}$. Calculate susceptibility at $300\ \mathrm{K}$. | **3** |
| 3 | What is the Néel temperature? Explain qualitatively what happens to antiferromagnetic order when a material is heated through this temperature. | **3** |
| 4 | Briefly explain the origin of ferrimagnetic moments. | **3** |
| 5 | Define electric polarisation $\mathbf{P}$ and electric susceptibility $\chi_e$. Write their relation to the macroscopic electric field in a linear isotropic dielectric, and state the SI unit of $\mathbf{P}$. | **3** |
| 6 | What is the local electric field at an atom? State the Lorentz local-field relation for an atomic site of cubic symmetry, and distinguish this field from the macroscopic electric field. | **3** |
| 7 | Define atomic polarisability and state its SI unit. Explain why polarisability is an atomic property whereas the dielectric constant is a bulk property. | **3** |
| 8 | Name the three principal sources of polarisability in a dielectric. Which contribution normally remains at optical frequencies, away from absorption resonances? | **3** |
| 9 | Discuss the ferroelectric and paraelectric states. | **3** |
| 10 | Explain the direct and inverse piezoelectric effects. Give one example of a piezoelectric material. | **3** |
| | **Section B** | |
| | **All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)** | |
| 11 | Describe the Langevin's theory of paramagnetism and obtain an expression for paramagnetic susceptibility. Mention the temperature dependence of susceptibility. | **6** |
| 12 | Derive Curie-Weiss law for ferromagnetism. | **6** |
| 13 | Explain the two-sublattice model of an antiferromagnet with a labelled sketch. Describe qualitatively the temperature dependence of the susceptibility parallel and perpendicular to the ordered moments below the Néel temperature, and the behaviour above it. No molecular-field algebra is required. | **6** |
| 14 | Explain ferrimagnetism using magnetite, $\mathrm{Fe_3O_4}$, as an example. Describe the occupation of the two magnetic sublattices and explain which ionic moments cancel and which give the net moment. Why can the material exhibit spontaneous magnetisation and hysteresis? | **6** |
| 15 | For a linear isotropic dielectric, derive $\mathbf{D} = \varepsilon_0(1+\chi_e)\mathbf{E}$. A sample has $P = 3.54 \times 10^{-6}\ \mathrm{C\,m^{-2}}$ in a macroscopic field $E = 2.00 \times 10^{5}\ \mathrm{V\,m^{-1}}$. Find $\chi_e$, $\varepsilon_r$ and $D$. Take $\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F\,m^{-1}}$. | **6** |
| 16 | Derive the Lorentz local field $\mathbf{E}_{\mathrm{loc}} = \mathbf{E} + \mathbf{P}/(3\varepsilon_0)$ for an atom of cubic symmetry, explaining each term. | **6** |
| 17 | Describe macroscopically the origin of static dielectric constant of a substance and discuss the classical theory of electronic polarization. | **6** |
| 18 | Explain piezoelectricity in terms of ionic displacement under stress. Why is the absence of a centre of inversion necessary for this effect? Give an example of a crystal that is piezoelectric but not ferroelectric, and explain one practical application of the direct or inverse effect. | **6** |
| | **Section C** | |
| | **Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)** | |
| 19 | Derive the Clausius–Mossotti relation $\dfrac{\varepsilon_r - 1}{\varepsilon_r + 2} = \dfrac{N\alpha}{3\varepsilon_0}$ from the local field. Explain the three sources of polarisability and how their contributions change with frequency. | **10** |
| 20 | (a) Explain spontaneous and switchable polarisation in a ferroelectric. Sketch its polarisation–electric field hysteresis loop and identify remanent polarisation and coercive field. **(4 marks)** (b) Explain qualitatively how the relative positions of positive and negative ions in barium titanate change across the cubic-to-tetragonal transition and produce a ferroelectric state. **(4 marks)** (c) Explain why a piezoelectric crystal need not be ferroelectric, using a suitable example. **(2 marks)** | **10** |